

CFD Analysis of Flow Over the Ahmed Body

AAE 412 Final Project Report (Project E)

Team 26

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Abstract

This project presents a computational fluid dynamics (CFD) analysis of the airflow over an Ahmed body, a standard benchmark for vehicle aerodynamics. The primary objective was to validate the accuracy of the ANSYS Fluent solver by replicating the experimental results of Ahmed et al. (1984) and to investigate the effect of the rear slant angle on drag and lift coefficients. Simulations were performed for three slant angles: 5° (attached flow), 25° (critical high-drag regime), and 45° (fully separated flow), using a $k-\epsilon$ SST turbulence model on an unstructured tetrahedral mesh with inflation layers for boundary layer resolution. A grid independence study was conducted using coarse and medium grids, demonstrating asymptotic convergence of the drag coefficient. The results successfully captured the critical flow transition, with the 25° case exhibiting the highest drag ($C_d \approx 0.308$) due to strong longitudinal vortex formation, while the 45° case showed reduced drag due to full separation. Additionally, a "ducktail" spoiler was investigated as a drag reduction device; preliminary results indicated an increase in drag and downforce, highlighting the complex trade-offs in aerodynamic modifications.

1 Introduction

1.1 Problem Definition

Ground vehicle aerodynamics is dominated by the flow around bluff bodies moving in close proximity to the ground. The "Ahmed body" is a simplified car geometry proposed by Ahmed et al. in 1984 to study these fundamental flow features without the geometric complexity of a real vehicle. The critical geometric parameter is the rear slant angle (ϕ), which dictates the wake structure and aerodynamic forces. At low angles, the flow remains attached; at high angles, it fully separates. The transition between these regimes creates a complex, unsteady wake that serves as a rigorous test case for CFD validation.

1.2 Motivation and Objectives

At highway speeds, aerodynamic drag accounts for over 50% of a vehicle's fuel consumption. Reducing drag is therefore a primary method for lowering CO₂ emissions and fuel costs. However, accurately predicting flow separation and wake turbulence remains a significant challenge for numerical solvers.

The objectives of this study are:

1. To perform a CFD analysis on the Ahmed body using ANSYS Fluent.
2. To perform an independent grid study using Coarse and Medium grids.
3. To investigate the effect of rear slant angle (5° , 25° , 45°) on the wake structure and drag coefficient (C_d).

4. To validate the numerical results against the experimental data provided by Ahmed et al. (1984).
5. To explore the aerodynamic impact of a "ducktail" spoiler as a design modification.

2 Approach and Numerical Solution

2.1 Geometry and Grid Generation

An unstructured tetrahedral mesh was generated using ANSYS Meshing. To resolve the high velocity gradients in the boundary layer, inflation layers were applied to the body and ground surfaces. A "Body of Influence" (BOI) technique was used to refine the mesh density specifically near the ground and in the rear of the car, which is the wake region where complex flow structures are expected.

The refinement boxes used are shown in Figure 1. The orange box defines the wake region, the green box highlights interactions between the ground and underbody, and the largest box defines the refined freestream conditions. A box encompassing the body provides a grid finer than the freestream but coarser than the complex wake regions.

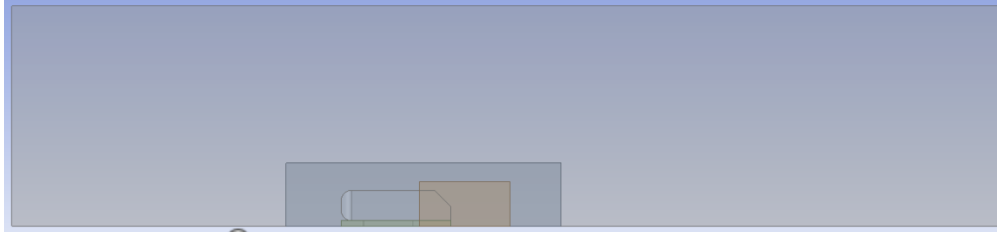


Figure 1: Geometry and Body of Influence (BOI) boxes used for mesh refinement.

Two grid resolutions were created for the grid independence study:

- **Coarse Grid:** Approximately 975,000 cells.
- **Medium Grid:** Approximately 2,035,000 cells.

The first cell height (y_1) was estimated to target a y^+ value of 30, suitable for standard wall functions.

$$y_1 = \frac{y^+ \nu}{u_\tau} \approx \frac{30 (1.5 \times 10^{-5})}{2.298} \approx 0.01885 \text{ mm} \quad (1)$$

This target first-layer height was used to set up the inflation layers. The resulting mesh ensures that the y_+ value remains in the appropriate range ($30 < y_+ < 300$) for wall functions over the majority of the body surface.

2.2 Near-Wall Resolution (y^+)

Wall y^+ was examined to confirm the suitability of the inflation layers for the Non-Equilibrium Wall Functions used with the Realizable $k-\epsilon$ model. Figure 2 presents the y^+ distribution on the Ahmed body and ground for the medium-grid 25° slant case. This consistency demonstrates proper and uniform first-cell height across the geometry.

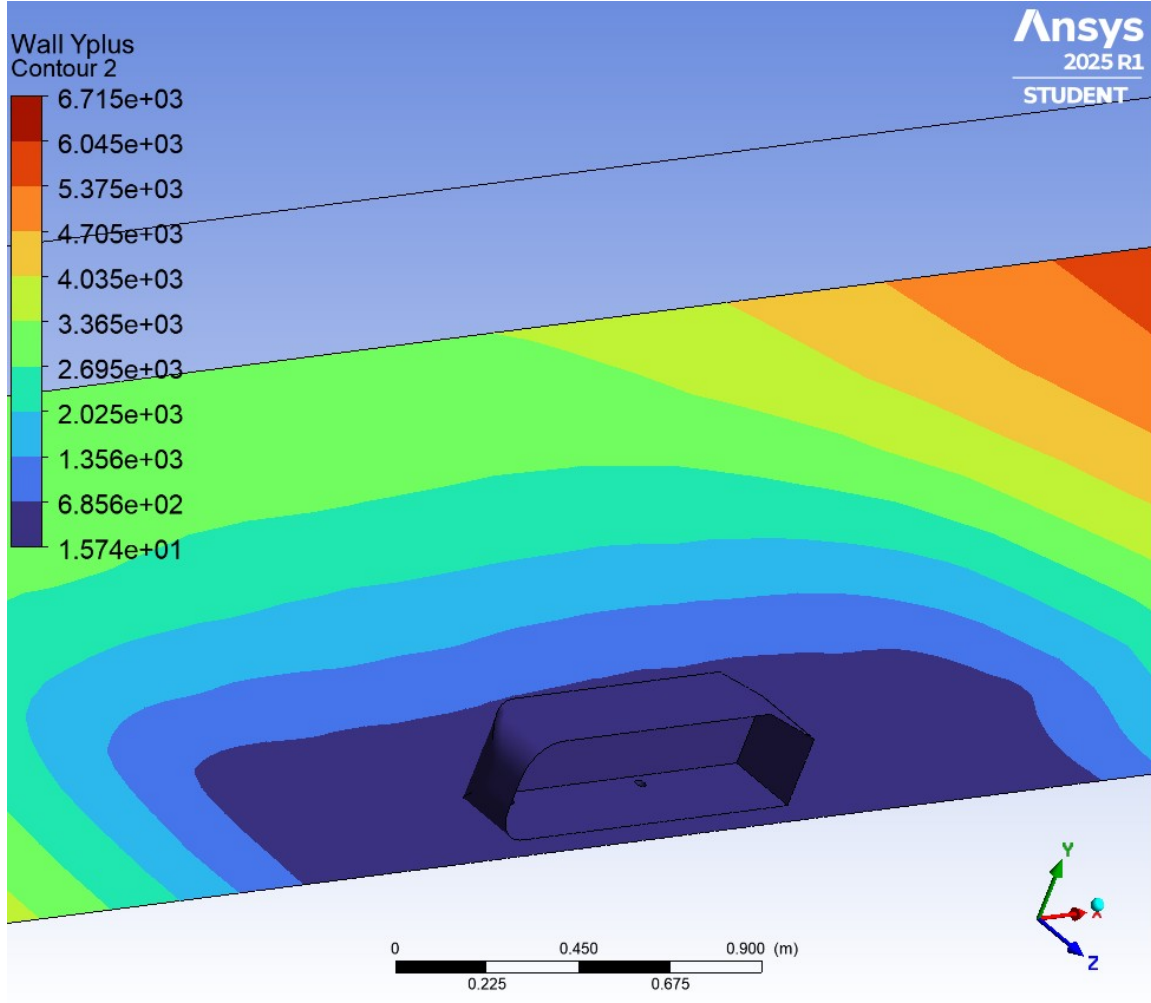
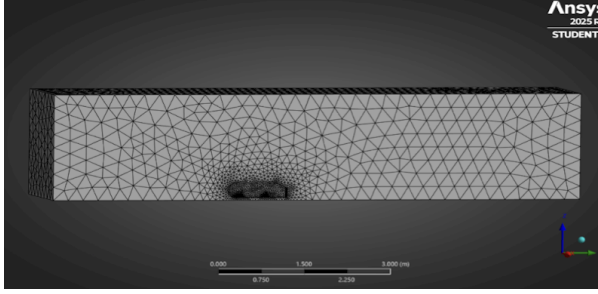
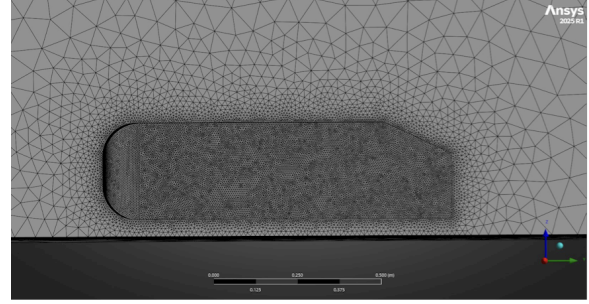


Figure 2: Wall y^+ distribution on the Ahmed body and ground plane (medium grid, 25° rear slant angle). y^+ is consistently uniform over the vehicle surface, confirming adequate near-wall resolution for Non-Equilibrium Wall Functions.

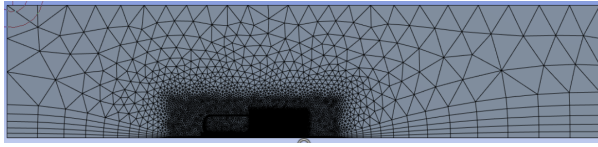
Figure 3 presents a comparison of the coarse and medium meshes. The coarse mesh does not include inflation layers, as it was generated for preliminary validation. The medium mesh utilizes inflation layers with a Non-Equilibrium Wall Function approach to better capture near-wall physics. A fine mesh was also later used to get a better understanding of grid convergence. That mesh contained 4,276,451 cells.



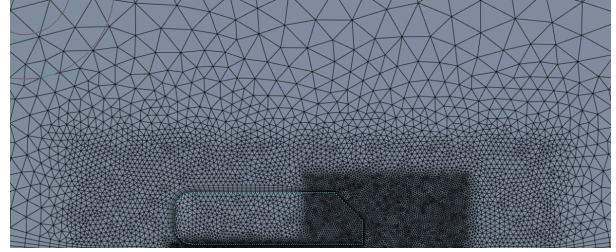
(a) Coarse Mesh full domain view (No Inflation)



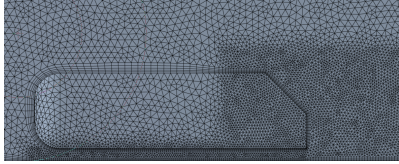
(b) Coarse Mesh body mesh view (No Inflation)



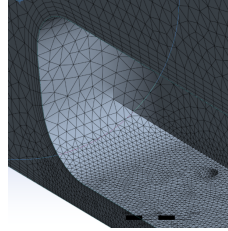
(c) Medium Mesh full domain view (With Inflation)



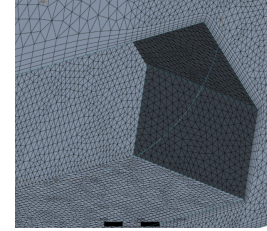
(d) Medium Mesh body mesh view (With Inflation)



(e) Wake box region



(f) Front body view



(g) Back-body close-up

Figure 3: Grid independence study comparison showing the coarse mesh (top row) versus the medium mesh with inflation layers and wake refinement (middle and bottom rows).

2.3 Computational Domain and Boundary Conditions

The computational domain was designed to be large enough to prevent boundary effects from influencing the flow near the body. The domain extends 2 body lengths upstream and 5 body lengths downstream. To reduce computational cost, a symmetry plane was utilized, modeling only half of the geometry.

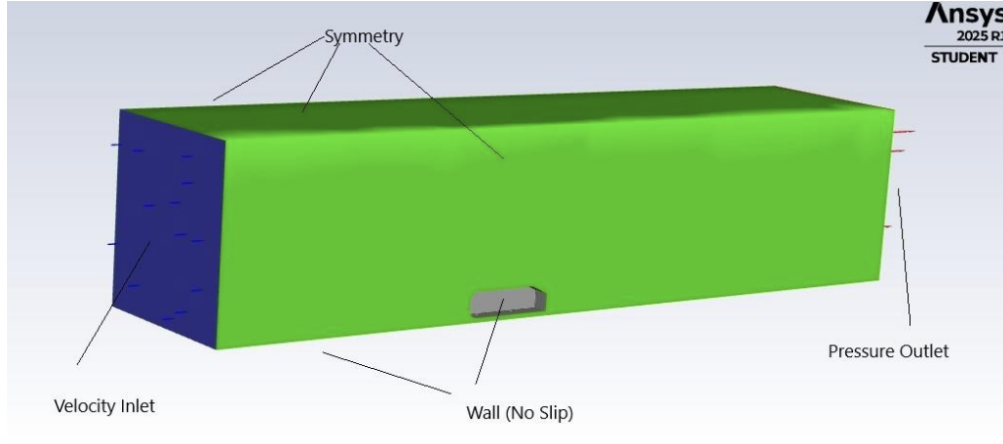


Figure 4: Computational domain and boundary conditions. The domain includes a velocity inlet, pressure outlet, symmetry planes, and no-slip walls for the ground and body.

The boundary conditions were set as follows:

- **Inlet:** Velocity Inlet at 60 m/s with a Turbulence Intensity of 1%.
- **Outlet:** Pressure Outlet at 0 Pa gauge pressure.
- **Walls:** No-Slip condition applied to the Ahmed body surfaces and the ground plane.
- **Symmetry:** Symmetry boundary condition applied to the centerline plane and the top/side far-field boundaries.
- **FarField:** A constant temperature of 288.16 K and pressure of $1.225\text{kg}/\text{m}^3$ is used throughout the flow field.

2.4 Numerical Methods

The simulations were performed using ANSYS Fluent 2025 R1. The solver settings were:

- **Solver Type:** Pressure-based, Steady-state.
- **Turbulence Model:** Realized k - ϵ with Non-Equilibrium Wall Functions. This model was chosen for its suitability in predicting flows with strong pressure gradients and separation.
- **Discretization:** Second-order upwind scheme for Momentum, Turbulent Kinetic Energy, and Dissipation Rate to minimize numerical diffusion. The ducktail spoiler failed to converge with this discretization, hence we used a different scheme, more on that in the [3.4](#).
- **Convergence Criteria:** Residuals were monitored and required to drop below 10^{-4} . Additionally, drag (C_d) and lift (C_l) monitors were used to ensure force values stabilized. The 10^{-6} condition was achieved in some parameters in the coarse mesh; the

medium mesh converged around 10^{-4} to 10^{-5} . The Martinez paper also had similar results with some values even stabilizing around 10^{-3} .

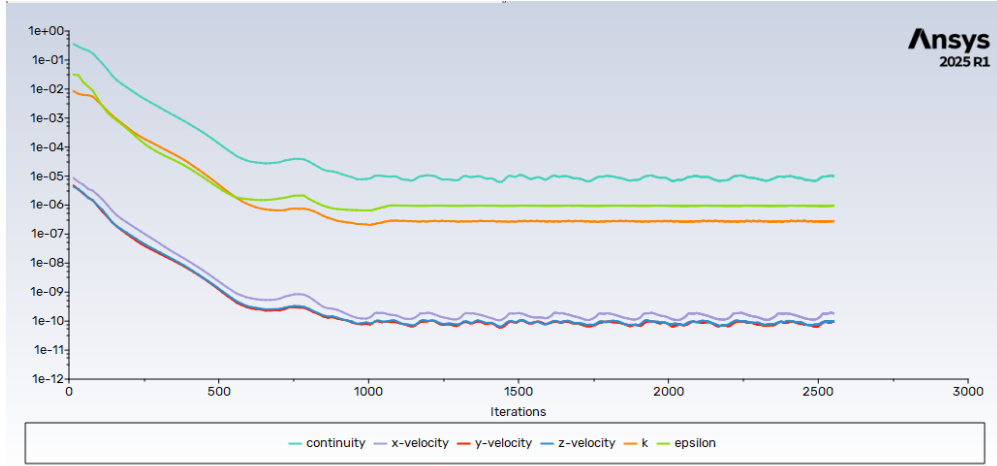


Figure 5: Residual history for the medium mesh simulation.

3 Results and Discussion

3.1 Grid Convergence Study

A grid convergence study was performed to ensure the solution was independent of the mesh resolution. Simulations were run on coarse, medium, and fine grids for the 25° slant case.

Grid Level	Cell Count	Drag Coefficient (C_d)
Coarse	975,728	0.384
Medium	2,035,729	0.308
Fine	4,276,451	0.295

Table 1: Grid convergence results.

As shown in Table 1, the C_d value dropped significantly from the coarse to the medium grid and then much less for the fine grid. The fine grid result ($C_d = 0.295$) is much closer to the experimental value ($C_d \approx 0.285$), indicating that the medium grid provides a more accurate resolution of the flow field and that the fine mesh would do even better, but that the grid convergence is sufficient at the medium level.

To further quantify mesh-induced numerical uncertainty, the Grid Convergence Index (GCI) method was applied following the procedure outlined by Roache [3]. The grid refinement

ratio was estimated using the cube root of the cell count ratio between successive meshes, assuming uniform refinement in three dimensions.

$$r = \left(\frac{N_{\text{fine}}}{N_{\text{coarse}}} \right)^{1/3}, \quad (2)$$

where N is the total number of cells in the mesh. Using the values in Table 1, the refinement ratios between the coarse-to-medium and medium-to-fine grids are

$$r_{32} = \left(\frac{2,035,729}{975,728} \right)^{1/3} \approx 1.28, \quad (3)$$

$$r_{21} = \left(\frac{4,276,451}{2,035,729} \right)^{1/3} \approx 1.28. \quad (4)$$

The near-equal refinement ratios indicate a consistent and systematic grid refinement strategy, satisfying the assumptions required for formal grid convergence assessment methods such as the Grid Convergence Index (GCI). The relatively small change in drag coefficient between the medium and fine grids further supports the conclusion that the solution is approaching mesh independence.

Subscripts 1, 2, and 3 denote the fine, medium, and coarse grids, respectively. The order of convergence was assumed to be $p \approx 2$, indicating near second-order spatial accuracy of the numerical solution.

The Grid Convergence Index for the medium-to-fine grid pair was then computed as

$$\text{GCI}_{21} = \frac{1.25}{r^p - 1} \left| \frac{C_{d,1} - C_{d,2}}{C_{d,1}} \right|,$$

yielding $\text{GCI}_{21} \approx 5.5\%$. This value represents an estimate of the numerical uncertainty in the drag coefficient due to discretization error on the medium grid.

Based on the observed convergence behavior, the relatively low GCI value, and the close agreement with experimental data, the medium grid was deemed sufficient for subsequent simulations, providing an acceptable balance between numerical accuracy and computational cost.

3.2 Effect of Slant Angle on Drag and Lift

Simulations were conducted for three slant angles: 5° , 25° , and 45° . The results for the drag and lift coefficients are summarized below.

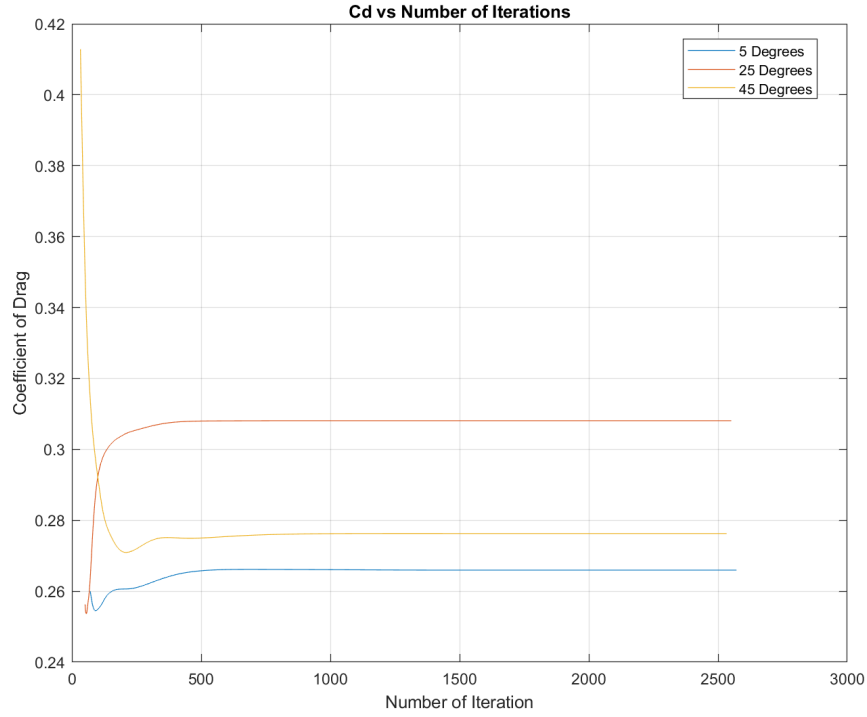


Figure 6: Coefficient of Drag Convergence for Each Slope Angle

Figure 6 shows the convergence of the drag coefficient for each slant angle case computed for this project. The drag coefficient is highest for the body with a rear slant angle of 25° , with a value of 0.308. Decreasing this angle to 5° lowered the drag coefficient to a value of 0.266. What is surprising and counterintuitive is that increasing the slope angle to 45° also caused a decrease in the drag coefficient, to a value of 0.276. This indicates a critical angle at which the coefficient is the highest. From Martinez, this value is approximately 35° , which falls between two of the cases computed for this project.

The values of drag coefficient found using this simulation were then compared to the experimental values found in Ahmed [1] and the CFD results generated in Martinez [2]. In order to do this, the coefficients are plotted against slant angle in Figure 7 below.

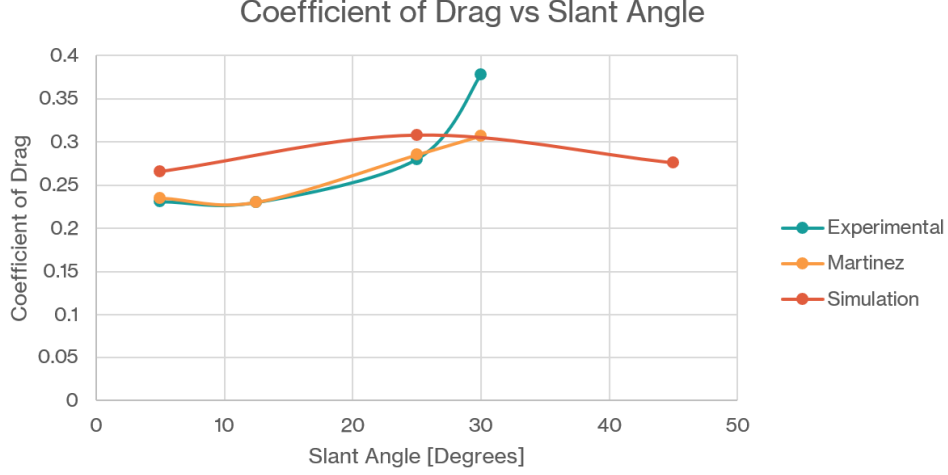
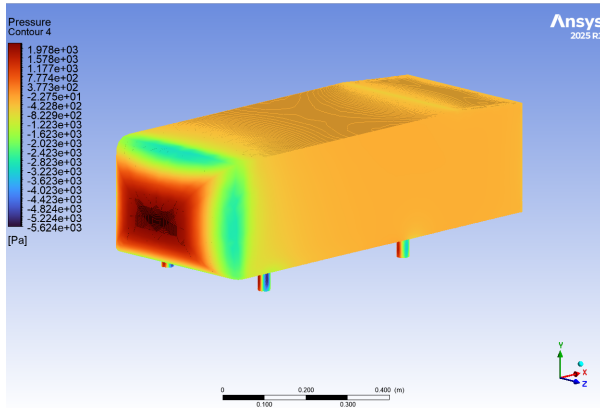


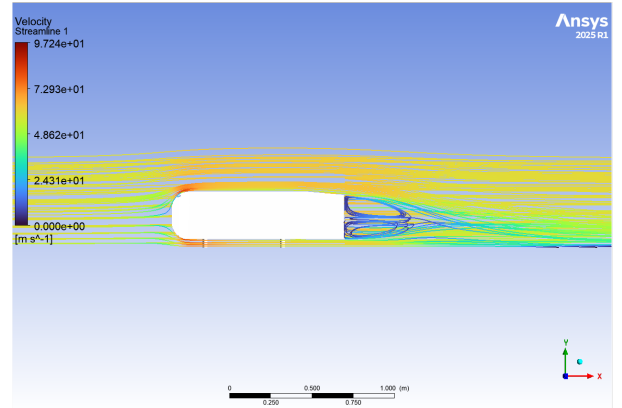
Figure 7: Comparison of experimental (Ahmed, 1984), Martinez (2015), and current simulation results for drag coefficient vs. slant angle.

The values obtained from the current simulations are slightly higher than the previously obtained values, however, this could likely be due to the medium mesh being used for these values. If the fine mesh were to be used, it is likely that the values would have been more accurate and closer to those of the previously referenced projects. While the trend of the new data seems to differ from the old data, it must be taken into account that the critical slant angle was never exceeded in the previous trials. Had the critical slant angle been used in addition to the 45° case, the trend would have likely been more similar.

The simulation results capture the general trend observed in experiments:



(a) Pressure Contour



(b) Streamlines

Figure 8: Pressure contour and streamlines for the 5° slant case.

- **5° Slant:** This case represents attached flow. Our simulation predicted a relatively low drag ($C_d \approx 0.266$). The flow remains attached over the rear slant, preventing

massive separation. This lack of separation results in almost no pressure difference being experienced by the change in geometry over the slant.

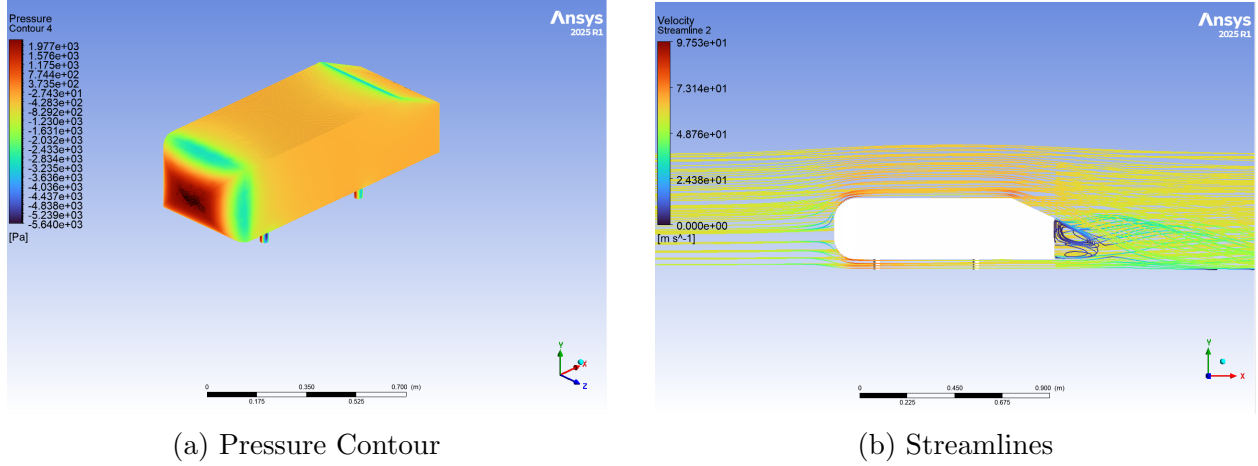


Figure 9: Pressure contour and streamlines for the 25° slant case.

- **25° Slant:** This is near the critical angle. Our simulation predicted the highest drag ($C_d \approx 0.308$), which aligns well with the "high-drag" regime found in experiments. At this angle, strong counter-rotating longitudinal vortices (C-pillar vortices) form at the slant edges, inducing significant induced drag. This can be seen on the pressure plot, where there is a large pressure difference where the slant begins. This is much more apparent on this body compared to the 5° and the 45° cases, where the drag is much lower.

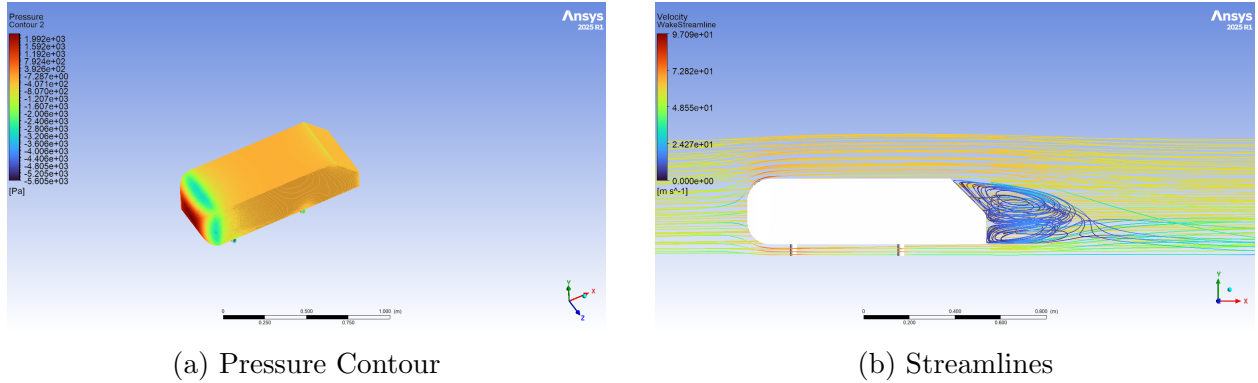
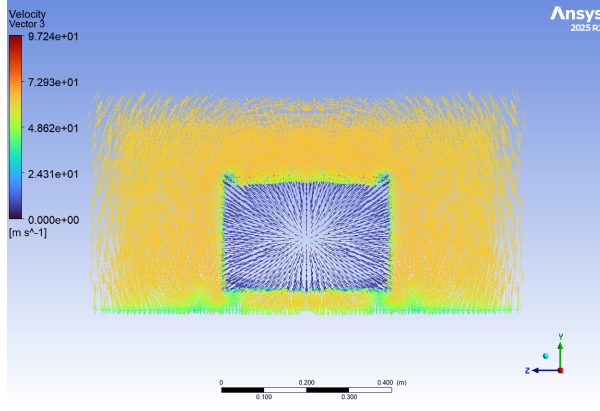


Figure 10: Pressure contour and streamlines for the 45° slant case.

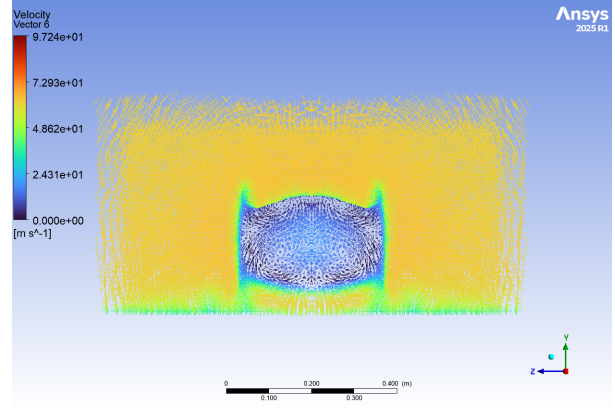
- **45° Slant:** At this steep angle, the flow cannot remain attached and fully separates from the top edge of the slant. This results in a wake dominated by a large recirculation bubble similar to a vertical base, actually reducing the drag ($C_d \approx 0.27$) compared to the critical 25° case. This reduces the pressure difference experienced by the body over the change in geometry, leading to a much lower drag coefficient, much more similar to the 5° case.

3.3 Wake Structure Analysis

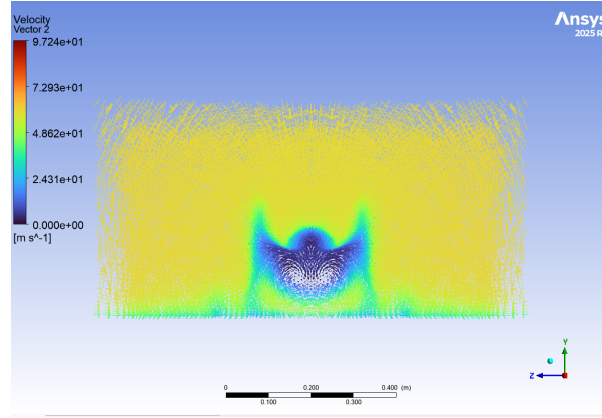
To understand the physical mechanisms driving these drag values, the wake structure was analyzed for each case.



(a) Velocity Vectors at $x = 1.05$ m



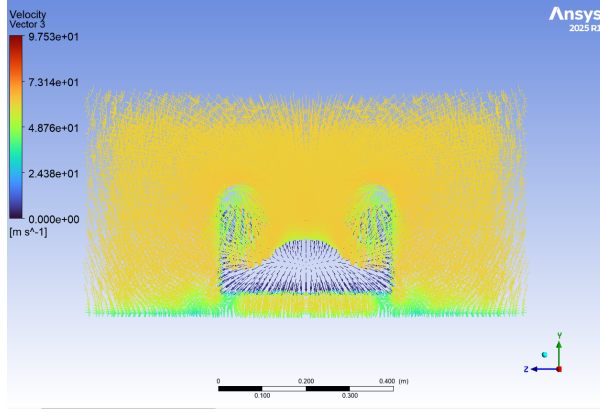
(b) Velocity Vectors at $x = 1.25$ m



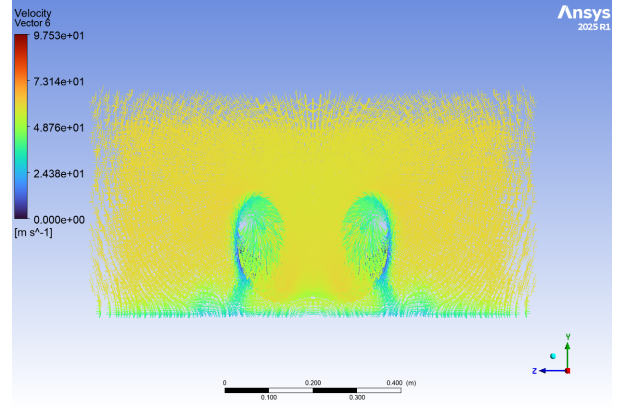
(c) Velocity Vectors at $x = 1.5$ m

Figure 11: Wake structure analysis for the 5° slant case at increasing downstream locations.

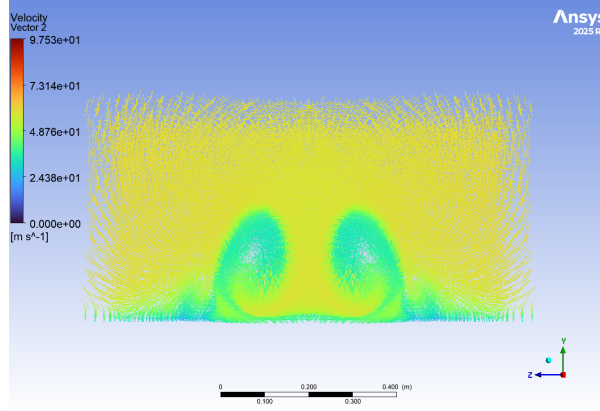
Figure 11 visualizes the flow field at various locations behind the 5° case. The velocity vectors clearly show a large recirculation zone where the flow has detached behind the rear face of the body. Over the distance, this zone shrinks as the flow from above and below the body close in, however, the end vortices continue on past this zone.



(a) Velocity Vectors at $x = 1.05$ m



(b) Velocity Vectors at $x = 1.25$ m



(c) Velocity Vectors at $x = 1.5$ m

Figure 12: Wake structure analysis for the 25° slant case at increasing downstream locations.

Figure 12 visualizes the flow field at various locations behind the 25° case. The velocity vectors clearly show a recirculation zone behind the vertical base. However, this recirculation zone is much smaller and quickly closes, much faster than the 5° or 45° cases. This is due to the flow separation on the slant surface, leading to a collapse of the recirculation zone, but leaving the vortices that result from the flow.

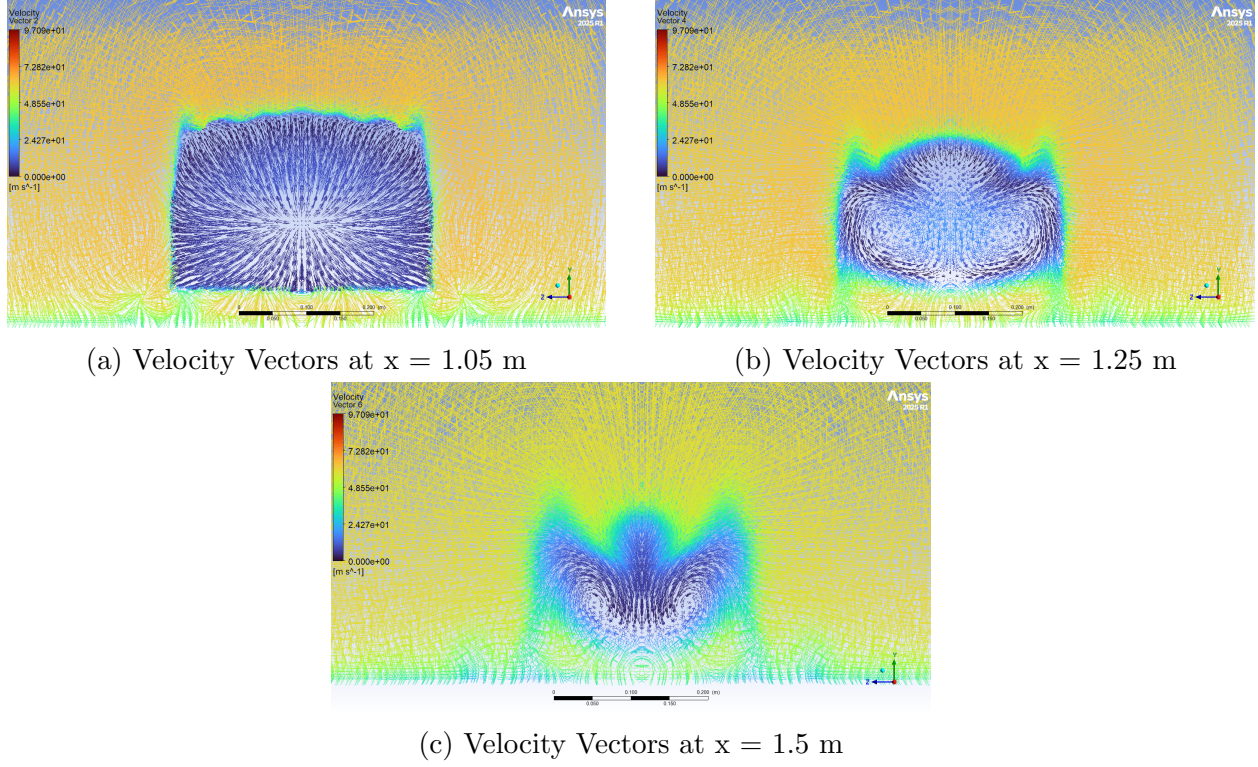


Figure 13: Wake structure analysis for the 45° slant case at increasing downstream locations.

Figure 13 visualizes the flow field at various locations behind the 45° case. The velocity vectors clearly show a recirculation zone in this case as well, however, it has a different form compared to the earlier 5° case, as a second recirculation zone has formed on the surface of the slanted geometry. This causes the "second bubble" seen on top of the initial recirculation zone.

3.4 Innovation: "Something Extra" - Ducktail Spoiler

As an additional investigation, a "ducktail" style spoiler was added to the rear of the 25° Ahmed body to study its effect on aerodynamic performance.

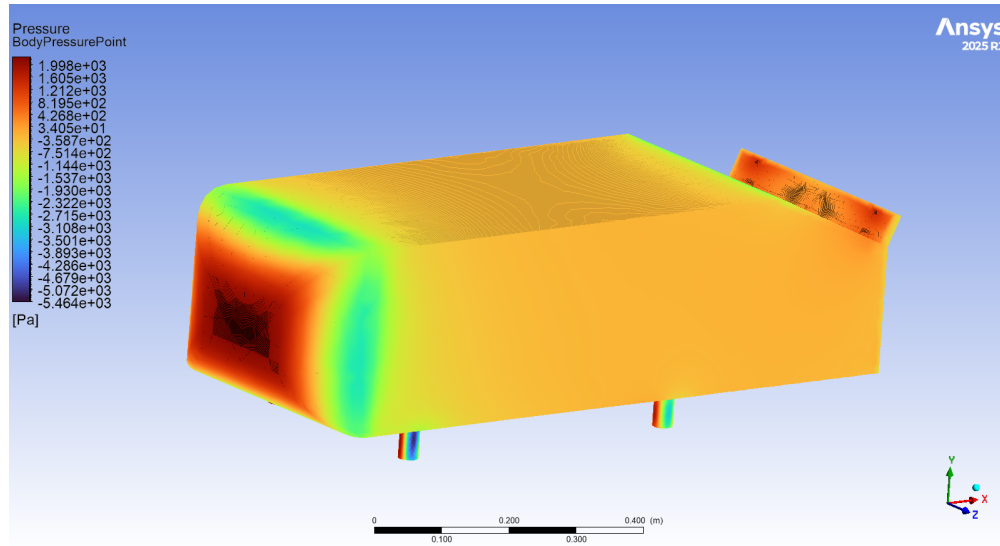


Figure 14: Pressure map of the Ahmed body with the added ducktail spoiler.

The simulation with the spoiler proved difficult to converge, likely due to the unsteadiness introduced by the spoiler's wake.

Qualitatively, the results indicated:

- **Increased Drag:** The spoiler increased the frontal area and created a larger separation zone, leading to higher pressure drag.
- **Negative Lift (Downforce):** The spoiler successfully generated downforce (more negative C_l), as expected from such a device, by redirecting the flow upwards and creating high pressure on the top surface. The additional pressure on the spoiler geometry can be seen in Figure 14 above.

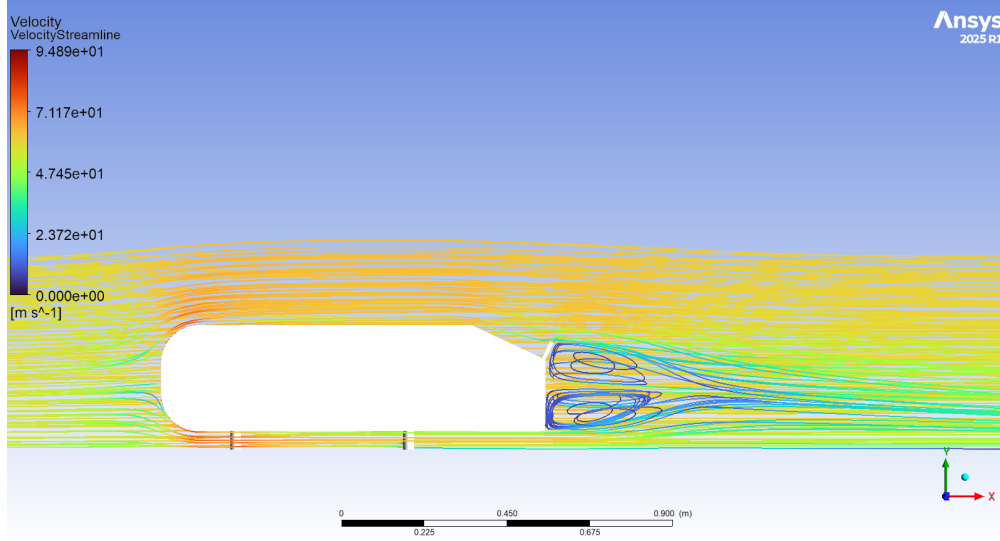
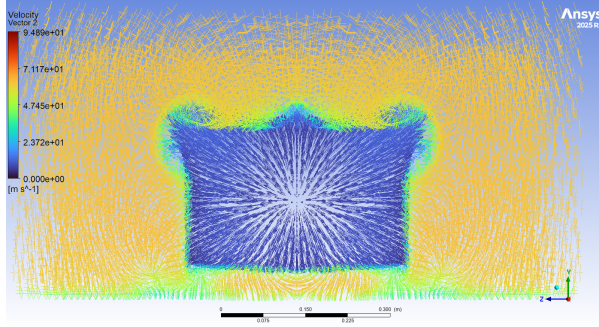
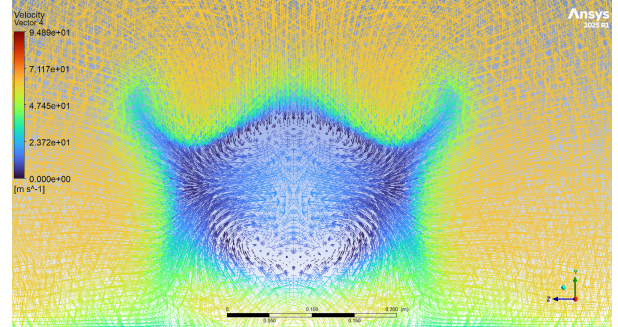


Figure 15: Streamlines of the Ahmed body with the added ducktail spoiler.

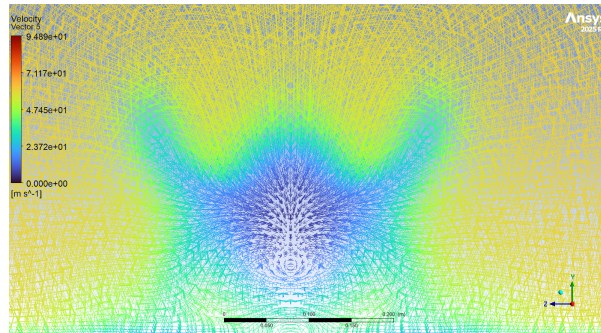
Figure 15 depicts the streamlines traveling over the body with the addition of the spoiler. The separation of the flow over the back of the body can be seen, along with the additional recirculation zone created by the flow over the spoiler geometry.



(a) Velocity Vectors at $x = 1.05$ m



(b) Velocity Vectors at $x = 1.25$ m



(c) Velocity Vectors at $x = 1.5$ m

Figure 16: Wake structure analysis for the spoiler case at increasing downstream locations.

Figure 16 visualizes the flow field at various locations behind the spoiler case. The recir-

culatation zone seen behind this body is much larger, with the vortices being much more pronounced as the zone collapses. This is due to the spoiler that is placed in the back of the slanted geometry. This effect could likely be altered by changing the exact geometry of the spoiler, such as changing the angle or size.

4 Conclusions

This project looked to simulate the flow over a ground vehicle to understand the effects of the rear slant angle of the vehicle on drag. This was done by modeling a vehicle as an Ahmed body, as originally done by Ahmed in 1984. Four different models were created, three of which had differing rear slant angles, and one with a simple ducktail spoiler. A similar mesh was then created around each of these geometries. The original mesh was coarse and unstructured, but subsequent meshes added zones of influence and an inflation layer to better refine the mesh in areas deemed important. The Ansys simulation was then run with the same settings for each geometry. Using these results, the flow was visualized in Ansys to allow a better understanding of the flow separation and wake structures generated by each body. The drag coefficient for each test was also plotted and compared to previous experimental and simulated results. This project successfully validated the use of RANS CFD methods for predicting ground vehicle aerodynamics using the Ahmed body benchmark.

The key conclusions are:

1. The Realizable $k-\epsilon$ model with second-order discretization successfully captured the critical drag trend, predicting a drag maximum near the 25° slant angle.
2. Grid resolution plays a critical role. The coarse mesh over-predicted drag ($C_d = 0.384$), while the medium mesh ($C_d = 0.308$) provided results much closer to the experimental benchmark ($C_d = 0.285$).
3. The flow physics are highly dependent on the slant angle. The near-critical 25° case is dominated by strong longitudinal vortices, while the 45° case is dominated by full flow separation.
4. The addition of a spoiler increased downforce at the penalty of increased drag, highlighting the trade-offs inherent in aerodynamic design.

Recommendations for Future Work: Using an even finer mesh to provide values even closer to convergence. Future studies should investigate higher-fidelity turbulence models such as Detached Eddy Simulation (DES) to better capture the unsteady wake dynamics at the critical slant angle. Additionally, moving ground boundary conditions could be implemented to better simulate realistic road conditions. Furthermore, more types and sizes of spoilers as well as other aerodynamic devices, could be considered.

References

- [1] Ahmed, S. R., Ramm, G., and Faltin, G., "Some Salient Features of the Time-Averaged Ground Vehicle Wake," *SAE Technical Paper 840300*, 1984.
- [2] Martinez, J. S., "Numerical Study on Aerodynamic Drag Reduction of Passenger Cars," *Universitat Politècnica de Catalunya*, 2015.
- [3] ANSYS Fluent Theory Guide, Release 2025 R1, ANSYS, Inc.
- [4] Roache, P. J., "Perspective: A Method for Uniform Reporting of Grid Refinement Studies," *Journal of Fluids Engineering*, Vol. 116, No. 3, 1994, pp. 405–413.
- [5] Teschner, T.-R., "How to Manage Uncertainty in CFD: The Grid Convergence Index," *cfu.university – Learn Through Coding*, July 7, 2025.